INCH-POUND

A–A–59590/9A
12 August 2005
SUPERSEDING
A-A-59590/9
1 March 2002

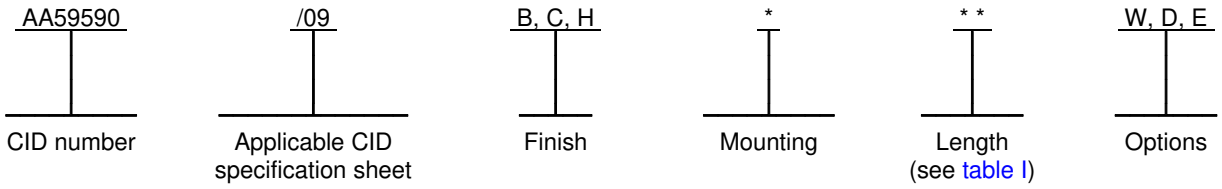
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE, WITH SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59590/09BV28E is the PIN for a hard black anodize finished, 2.8 inch (71.1 mm) long card holder with a retained actuating screw. The card holder also features two tapped mounting holes for use with 2-56 UNC 2B fasteners and a screw self-lock element for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces. The screw retention feature is to prevent unintentional disassembly of the card holder.

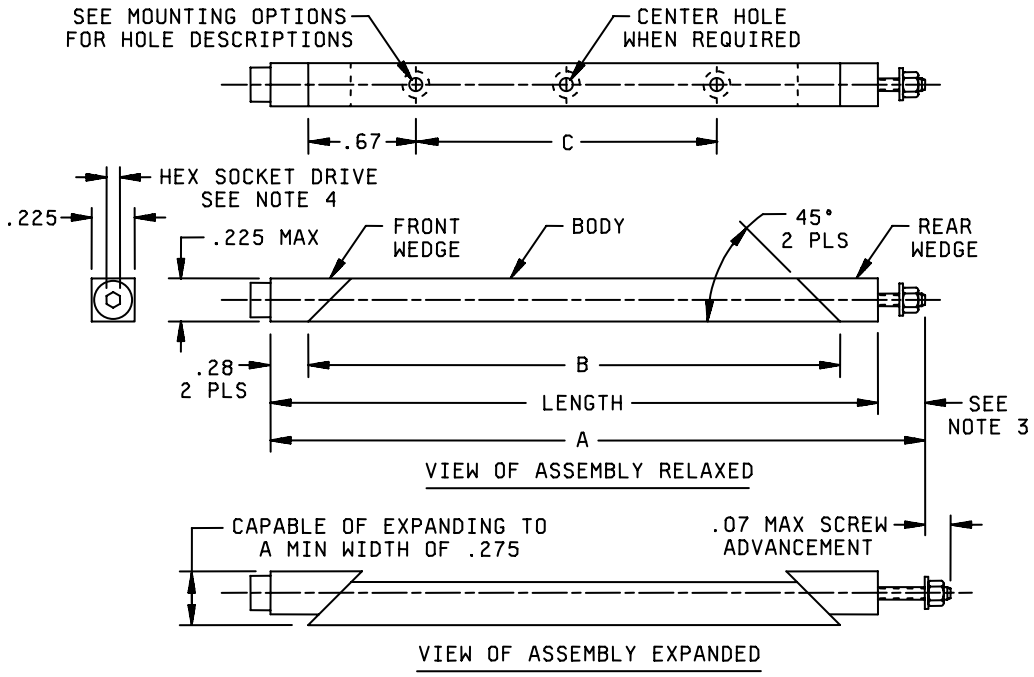
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table 1, and A–A–59590.

Screw retention (locknut). The screw retention feature intended to prevent unintentional disassembly of the card holder shall be as specified in figure 2. The locknut shall be fabricated of a corrosion resistant material.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

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Inches	mm	Inches	mm	Inches	mm
.07	1.8	.225	5.72	.28	7.1
.220	5.59	.275	6.99	.67	17.0

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.5 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders without option “W” or “D” and .30 (7.6 mm) maximum on card holders with option “W” or “D”.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (0.98 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.



FIGURE 2. Screw retaining self-locking nut (corrosion resistant) details.

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Finish. The finish options available are as follows: “B” (black anodize), “C” (chemical film), or “H” (hard black anodize) shall be as specified in A–A–59590.

Mounting. The mounting options shall be as specified in A–A–59590. The options available are as follows: “J” or “K” (rivet mounting holes), “N” (no mounting holes), “T” or “R” (tapped 0-80 UNF 2B holes), “V” or “S” (tapped 2-56 UNC 2B holes), or “G” or “M” (tapped M2.5 x.45 metric holes). See table I and figure 1 for mounting hole spacing requirements. Card holders using mounting option “J” or “K” may be shipped unassembled.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to 4.45 mm) deep.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths are listed in table I.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension “LENGTH” ± .02 (0.5)	Dimension “A” (maximum)	Dimension “B” ± .03 (0.8)	Dimension “C” ± .005 (0.13)
18	1.8 (45.7)	2.20 (55.9)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.70 (68.6)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.20 (81.3)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.40 (86.4)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.70 (94.0)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.90 (99.1)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.20 (106.7)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.70 (119.4)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.20 (132.1)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.70 (144.8)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.20 (157.5)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.70 (170.2)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.20 (182.9)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.70 (195.6)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.20 (208.3)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.70 (221.0)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.20 (233.7)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Options. Card holders can have the following options. Card holders not requiring the following option shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

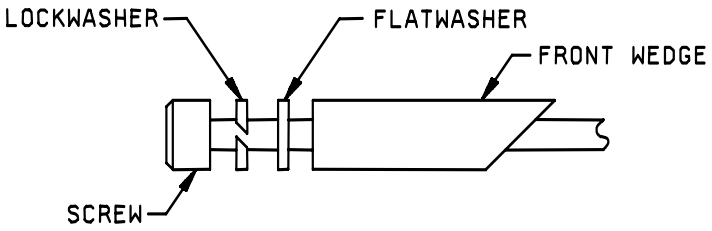


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	APW Electronic Solutions 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: sales.electronicsolutions@apw.com Uniform Resource Locator (URL): www.apw.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers as shown in table II. The CID PINs listed in table II are only for length designator "28". See table III for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	42–6–LN	CJ28	230CR2.80HR
CN28	42–6–LN–A	CN28	230CR2.80R
CT28	42–6–LN–T	CT28	230CR2.80T0R
CV28	42–6–LN–S	CV28	230CR2.80T2R
CG28	42–6–LN–M	CG28	M230CR2.80TM2.5R
CJ28E	42–6–LN–L	CJ28E	230CR2.80HLR
CN28E	42–6–LN–A–L	CN28E	230CR2.80LR
CT28E	42–6–LN–T–L	CT28E	230CR2.80T0LR
CV28E	42–6–LN–S–L	CV28E	230CR2.80T2LR
CG28E	42–6–LN–M–L	CG28E	M230CR2.80TM2.5LR
CJ28W	42–6–LF–LN	CJ28W	W230CR2.80HR
CN28W	42–6–LF–LN–A	CN28W	W230CR2.80R
CT28W	42–6–LF–LN–T	CT28W	W230CR2.80T0R
CV28W	42–6–LF–LN–S	CV28W	W230CR2.80T2R
CG28W	42–6–LF–LN–M	CG28W	MW230CR2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 52094
CJ28D	42–6–LF–LN–L	CJ28D	W230CR2.80HLR
CN28D	42–6–LF–LN–A–L	CN28D	W230CR2.80LR
CT28D	42–6–LF–LN–T–L	CT28D	W230CR2.80T0LR
CV28D	42–6–LF–LN–S–L	CV28D	W230CR2.80T2LR
CG28D	42–6–LF–LN–M–L	CG28D	MW230CR2.80TM2.5LR
BJ28	42–6–B–LN	BJ28	A230CR2.80HR
BN28	42–6–B–LN–A	BN28	A230CR2.80R
BT28	42–6–B–LN–T	BT28	A230CR2.80T0R
BV28	42–6–B–LN–S	BV28	A230CR2.80T2R
BG28	42–6–B–LN–M	BG28	MA230CR2.80TM2.5R
BJ28E	42–6–B–LN–L	BJ28E	A230CR2.80HLR
BN28E	42–6–B–LN–A–L	BN28E	A230CR2.80LR
BT28E	42–6–B–LN–T–L	BT28E	A230CR2.80T0LR
BV28E	42–6–B–LN–S–L	BV28E	A230CR2.80T2LR
BG28E	42–6–B–LN–M–L	BG28E	MA230CR2.80TM2.5LR
BJ28W	42–6–B–LF–LN	BJ28W	WA230CR2.80HR
BN28W	42–6–B–LF–LN–A	BN28W	WA230CR2.80R
BT28W	42–6–B–LF–LN–T	BT28W	WA230CR2.80T0R
BV28W	42–6–B–LF–LN–S	BV28W	WA230CR2.80T2R
BG28W	42–6–B–LF–LN–M	BG28W	MWA230CR2.80TM2.5R
BJ28D	42–6–B–LF–LN–L	BJ28D	WA230CR2.80HLR
BN28D	42–6–B–LF–LN–AL	BN28D	WA230CR2.80LR
BT28D	42–6–B–LF–LN–TL	BT28D	WA230CR2.80T0LR
BV28D	42–6–B–LF–LN–SL	BV28D	WA230CR2.80T2LR
BG28D	42–6–B–LF–LN–ML	BG28D	MWA230CR2.80TM2.5LR
HJ28	42–6–B3–LN	HJ28	HA230CR2.80HR
HN28	42–6–B3–LN–A	HN28	HA230CR2.80R
HT28	42–6–B3–LN–T	HT28	HA230CR2.80T0R
HV28	42–6–B3–LN–S	HV28	HA230CR2.80T2R
HG28	42–6–B3–LN–M	HG28	MHA230CR2.80TM2.5R

See footnote at end of table.

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TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ CAGE 52094
HJ28E	42–6–B3–LN–L	HJ28E	HA230CR2.80HLR
HN28E	42–6–B3–LN–A–L	HN28E	HA230CR2.80LR
HT28E	42–6–B3–LN–T–L	HT28E	HA230CR2.80T0LR
HV28E	42–6–B3–LN–S–L	HV28E	HA230CR2.80T2LR
HG28E	42–6–B3–LN–M–L	HG28E	MHA230CR2.80TM2.5LR
HJ28W	42–6–B3–LN–LF	HJ28W	WHA230CR2.80HR
HN28W	42–6–B3–LF–LN–A	HN28W	WHA230CR2.80R
HT28W	42–6–B3–LF–LN–T	HT28W	WHA230CR2.80T0R
HV28W	42–6–B3–LF–LN–S	HV28W	WHA230CR2.80T2R
HG28W	42–6–B3–LF–LN–M	HG28W	MWHA230CR2.80TM2.5R
HJ28D	42–6–B3–LF–LNL	HJ28D	WHA230CR2.80HLR
HN28D	42–6–B3–LF–LNAL	HN28D	WHA230CR2.80LR
HT28D	42–6–B3–LF–LNTL	HT28D	WHA230CR2.80T0LR
HV28D	42–6–B3–LF–LNSL	HV28D	WHA230CR2.80T2LR
HG28D	42–6–B3–LF–LNML	HG28D	MWHA230CR2.80TM2.5LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Example of PIN with available length designators.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BV18E	42–4–B–LN–S–L	BV18E	A230CR1.80T2LR
BV23E	42–5–B–LN–S–L	BV23E	A230CR2.30T2LR
BV28E	42–6–B–LN–S–L	BV28E	A230CR2.80T2LR
BV30E	3/	BV30E	A230CR3.00T2LR
BV33E	42–7–B–LN–S–L	BV33E	A230CR3.30T2LR
BV35E	3/	BV35E	A230CR3.50T2LR
BV38E	42–8–B–LN–S–L	BV38E	A230CR3.80T2LR

See footnotes at end of table.

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TABLE III. Example of PIN with available length designators.

PIN designator AA59590/09	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/09	Vendor similar designator or type part number 1/ 2/ CAGE 52094
BV43E	42–9–B–LN–S–L	BV43E	A230CR4.30T2LR
BV48E	42–10–B–LN–S–L	BV48E	A230CR4.80T2LR
BS53E	42–11–B–LN–S–L	BS53E	A230CR5.30ET2LR
BS58E	42–12–B–LN–S–L	BS58E	A230CR5.80ET2LR
BS63E	42–13–B–LN–S–L	BS63E	A230CR6.30ET2LR
BS68E	42–14–B–LN–S–L	BS68E	A230CR6.80ET2LR
BS73E	42–15–B–LN–S–L	BS73E	A230CR7.30ET2LR
BS78E	42–16–B–LN–S–L	BS78E	A230CR7.80ET2LR
BS83E	42–17–B–LN–S–L	BS83E	A230CR8.30ET2LR
BS88E	42–18–B–LN–S–L	BS88E	A230CR8.80ET2LR

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–008

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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